

That opportunity is already beginning to draw interest from India's technology services industry, particularly around quantum-safe communications, cybersecurity, and enterprise applications.

For instance, Tech Mahindra has been working on quantum-safe communications since 2018, including implementing quantum-secured communication over a 100-km optical fibre network. "Today, we are working on developing algorithms for post-quantum cryptography as well as quantum machine learning applications across sectors such as banking, high-frequency trading, and advanced image processing," says Nikhil Malhotra, Chief Innovation Officer and Global Head of AI & Emerging Tech, Tech Mahindra.

Although Indian start-ups are beginning to make progress, their scale remains far smaller than those in the US or China. India has more than 50 active quantum companies or start-ups, says Devroop Dhar, MD and CEO, Primus Partners. "The US is estimated to have more than 400 companies and China over 100 state-backed entities." The patent gap is even wider,

interest in building stronger linkages with Indian institutions through its broader network of quantum innovation hubs.

TALENT TURNAROUND

Talent is emerging as one of the most critical battle-grounds in the global quantum race. Unlike conventional software development, quantum technologies sit at the intersection of physics, mathematics, computer science, materials engineering, and advanced hardware systems, making specialised expertise far harder to build at scale.

India has no shortage of experts in mathematics, physics, computer science, and engineering. "The key gap is not theoretical capability. It is interdisciplinary engineering depth spanning cryogenics, control electronics, fabrication, error correction, photonics, cybersecurity, and applied quantum software," says Bhandarkar. He adds India needs to build an entire quantum workforce pyramid, with researchers at the top, specialised quantum engineers in the middle, and quantum-literate enterprise professionals at scale.

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with the US filing more than 100,000 quantum-related patents and China over 56,000; India is in triple digits. "However, investor interest is rising sharply, with Indian quantum start-ups raising more than \$40 million between January and October 2025, a 250% increase over the same period in 2024," he adds.

States too have started to compete for leadership in the sector. Andhra Pradesh is positioning Amaravati as a future Quantum Valley and partnering with IBM to bring an IBM Quantum System Two to the state. Karnataka, meanwhile, has announced a ₹1,000 crore Quantum Mission aimed at creating a \$20 billion quantum technology economy by 2035, alongside plans for a Quantum Hardware Park, prototyping facilities, and cloud-based Quantum-as-a-Service platforms.

International partnerships are also becoming central to India's strategy. India and Japan recently signed a Letter of Intent on cooperation in quantum science and technology spanning research, industrial deployment, and applications. Japan has also expressed

Recognising that talent development is central to the NQM, the government has started expanding quantum education across institutions. Undergraduate minor and MTech programmes in quantum technologies are being introduced. The government has also selected 23 institutions to establish quantum teaching labs. Singhee says more than 208,000 students enrolled this year in IBM's quantum computing MOOC launched in partnership with Indian Institute of Technology Madras.

But India should not try to win every part of the quantum race at once, says Mukhopadhyay. He adds India needs shared infrastructure and talent, including access to cryogenic systems, fabrication facilities, testbeds, and trained quantum engineers. Along with this, India should back a few selective hardware platforms rather than spreading funding across every possible architecture. The goal should be to build a deep quantum stack, not a thin layer of activity across too many areas.

India's opportunity may therefore lie less in trying to replicate the US or China and more in steadily building on its strengths. **BT**